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Experimental investigation on impact and spreading dynamics of a single ethanol–water droplet on a heated surface

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HIGHLIGHTS

- Impact and spreading dynamics of ethanol–water droplets are investigated.
- Low-We ethanol–water droplet shows better spreading than high-We water droplet.
- Lower surface tension and Marangoni effect affect the droplet impact.
- Impact regime maps are proposed with and without ethanol additive.

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ABSTRACT

In the present study, the impact and spreading dynamics of a single droplet on a heated surface are experimentally investigated. Ethanol-water droplets with ethanol concentrations of 0–8 vol% are adopted as the working fluid with a wide range of surface temperature from 75 °C to 450 °C and the Weber number ranges from 7 to 63. The instantaneous droplet evolution after impact is studied via visualization. All the observed phenomena implied that ethanol additive has a significant influence on both impact dynamics and spreading characteristics. A new phenomenon is observed that the addition of ethanol can compensate for the negative influence of small Weber numbers, *i.e.*, some ethanol–water droplets with smaller Weber numbers show better spreading dynamics, higher breakup and atomization possibility but weaker rebound compared with water droplets with higher Weber numbers. It is speculated that droplet impact is no longer solely dependent on the Weber number for ethanol–water droplet, and lower surface tension caused by ethanol additive and Marangoni effect due to concentration gradient along the interface are mainly responsible for this difference. The result also shows that impact behavior exhibits complex dependence on the surface temperature and Weber number, including deposition, rebound, rebound with atomization, breakup with atomization and breakup, while the regime map is significantly affected by ethanol additive.

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1. Introduction

1.1. Fundamentals of droplet impact

Droplet impact on a solid surface is extensively involved in various modern industrial and technological applications, such as spray quenching of metal alloys (Mascarenhas and Mudawar, 2010), spray cooling of electronic devices (Visaria and Mudawar, 2009), direct fuel injection in internal combustion engines (Panão and Moreira, 2005), plasma spraying coating (Cedelle et al., 2005), ink-jet printing (van Dam and Le Clerc, 2004) and steam turbine blade erosion (Ahmad et al., 2009). Although some of the processes are related to adiabatic impact, the dynamic behaviors and heat transfer of droplet impact on a heated surface are much more intricate but also fascinating. Problems of such a highly transient physical phenomenon were studied many decades ago and the investigation of such a process was subsequently conducted by many researchers. Despite the increasing applications and long-term analytical, numerical and experimental studies, the mechanisms of droplet impact are still far from well understood due to the inherent complexity of the phenomenon, and there is still much to do to acquire fundamental aspects of mass, momentum and heat transfer interactions during and after droplet impact.

Generally, the whole process of droplet impact on a heated surface is subject to a large number of factors simultaneously as fol-

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Nomenclature

a, b, c	constant [−]
c_p	constant-pressure specific heat [J/(kg·°C)]
d	droplet diameter [mm]
g	gravitational acceleration [m/s^2]
H	impact height [mm]
h_{fg}	latent heat of vaporization [J/kg]
Oh	Ohnesorge number [-]
Re	Reynolds number [-]
T	temperature [°C]
t	time [ms]
u	velocity [m/s]
We	Weber number [-]

Greek symbols

β	spreading factor [-]
ρ	density [kg/m^3]
σ	surface tension [mN/m]
μ	dynamic viscosity [mPa·s]
θ	static contact angle [°]

Subscripts

CHF	critical heat flux
LFP	Leidenfrost point
l	liquid
max	maximum
w	heating wall

lows: (1) droplet parameters, including the thermal-physical properties, *e.g.*, density, surface tension, dynamic viscosity and saturated temperature, as well as the impact velocity, the initial droplet size and temperature; (2) surface features, including the heating condition, wall material, thermal-physical properties, surface roughness, wettability and nano/micro structures; (3) ambient conditions, including the pressure, temperature, and the thermal-physical properties of the surrounding gas. It is advantageous to adopt dimensionless numbers to comprehensively integrate the effect of droplet parameters (Josserand and Thoroddsen, 2016). The Weber number ($We = \rho_l u^2 d/\sigma$) as the ratio of inertia force to the surface tension force, and the Reynolds number ($Re = \rho_l u d/\mu_l$) as the ratio of inertia force to the viscous force are generally used to analyze the droplet impact. It needs to mention that the initial droplet diameter and impact velocity component normal to the heated surface should be used. Additionally, the Ohnesorge number ($Oh = We^{1/2}Re^{-1}$) compares the viscous force to the inertial force as a function of only thermal properties.

1.2. Droplet impact behaviors

Among all these parameters mentioned in Section 1.1, the surface temperature and droplet Weber number might be two of the most significant factors from the perspectives of heat transfer and hydrodynamics, respectively. Purely depending on the surface temperature, the droplet after impact may undergo film evaporation, nucleate boiling, transition boiling or film boiling (Liang and Mudawar, 2017). Heat transfer mechanisms are entirely different for different modes, which in turn affects the hydrodynamic behaviors and vice versa. Therefore, the impact dynamics are even more complicated when coupling with heat transfer. Generally, the droplet may either deposit (droplet spreads and recedes but remains on the wall), splash (kinetic disintegration), rebound (droplet bounce by hydrophobicity or Leidenfrost effect), breakup (thermal disintegration) (Börnhorst and Deutschmann, 2018), and some complicated composite behaviors may also occur, *e.g.*, rebound with atomization and splash with atomization (Bertola, 2015). These behaviors are usually characterized by the so-called impact regime map which is constructed to demarcate various impact regimes over broad ranges of surface temperature and Weber number. A comprehensive summary of more detailed classifications of the impact patterns can be found in Ref. (Liang and Mudawar, 2017).

Although numerous experimental studies on droplet impact regime map were performed in the past few decades, the transition between different impact regimes are qualitative more or less due to the complicated phenomena observed during droplet impact on the heated surface, and different criteria are used by different authors. For example, some studies proposed that the boundary between rebound and rebound with atomization regimes corresponds to the temperature at the Leidenfrost point (T_{LFP}), which can be correlated as $T_{LFP} = a + bWe^c$, where the coefficients vary in different papers (Yao and Cai, 1988; Bertola and Sefiane, 2005; Bertola, 2015). Experimental results also indicated that deposition occurs during nucleate boiling, while rebound with atomization and breakup with atomization are hydrodynamic phenomena of transition boiling (Wang et al., 2019). Therefore, the temperature at critical heat flux (T_{CHF}) is regarded as a threshold for droplet deposition.

1.3. Alcohol additives effect

Although much research has been conducted to investigate the single-component droplet impact on a heated surface, there are much fewer experimental data and theory about bi-component droplet, not to mention the specified alcohol-added water droplet. Table 1 provides a summary of previous studies of alcohol additive effect on water droplet impact. These studies were conducted with various Reynolds and Weber numbers, initial droplet size, surface materials, as well as different heating conditions. However, a common element of these studies is that almost all the alcohol concentrations were relatively high.

As can be seen from the table, glycerol was mostly used in these studies. Water and glycerol have similar values of surface tension and density, but very different viscosity (*e.g.*, the viscosity of glycerol shows nearly a thousandfold increase than that of water at room temperature). Therefore, glycerol was generally adopted to study the viscosity effect on droplet impact. Other candidates, such as high contents of methanol, ethanol (Kudra et al., 1991; Marmanis and Thoroddsen, 1996; Marston et al., 2010) and isopropanol (Marmanis and Thoroddsen, 1996) with similar values of viscosity, whose surface tension are only a quarter to a third of that for water, were also added to pure water to observe the droplet impact (see Table 1), as well as sessile droplet experiments (Misyura, 2016; Sefiane et al., 2003) and molecular dynamics simulations (Lundgren et al., 2002; Surblys et al., 2014). Although the surface tension of 1-octanol, 2-ethyl-hexanol (Chen et al., 2018) and 2-butanol (Crooks and Boger, 2000) is also much smaller than water, the viscosity is several times higher than water. Therefore, the adoption of a small amount of ethanol in this experiment enables us to solely identify the surface tension effect, or equiva-

Table 1
Experimental studies of the alcohol additive effect on water droplet impact.

Reference	Alcohol type & concentration	Surface material	Test condition
Banks *et. al.* (Banks et al., 2014)	65–80 wt% glycerol	aluminum, Teflon and acrylic with hydrophilic coating	No heating, Re = 15–4000, We = 10–100, d_0 = 3.3 ± 0.2 mm
Bartolo *et. al.* (Bartolo et al., 2005)	glycerol	parafilm	No heating, u = 1–2.4 m/s, d_0 = 2.2–2.8 mm
Castanet *et. al.* (Castanet et al., 2015)	35–70 vol% glycerol	polished nickel	T_w = 600 °C, Re = 15–256, We = 3–160, d_0 = 0.1–0.3 mm
Chen *et. al.* (Chen et al., 2018)	100–1000 ppm 1-octanol, 2-ethyl-hexanol	sanded copper	T_w = 140–280 °C, We = 8–50, u = 0.5–1.1 m/s, d_0 = 1.8–2.3 mm
Cossali *et. al.* (Cossali et al., 2005)	8.8–60 wt% glycerol	aluminum alloy	T_w = 150–260 °C, u = 2.55–3.20 m/s, d_0 = 1.99–2.11 mm
Crooks & Boger (Crooks and Boger, 2000)	26–80% glycerol, 2-butanol	perspex and aluminum (polished and grooved), parafilm, dichlorosilane, polystyrene	No heating, H = 50–1000 mm, d_0 = 2.28–4.87 mm
Joung & Buie (Joung and Buie, 2014)	60–90% glycerol	porous titania	No heating, We = 327–402, d_0 = 2.9–3.0 mm
Kudra *et. al.* (Kudra et al., 1991)	20–75 wt% ethanol	stainless steel	T_w = 140–280 °C, We less than 80, d_0 = 0.4–0.8 mm
Laan *et. al.* (Laan et al., 2014)	50–76 vol% glycerol	stainless steel	No heating, Re = 70–17000, We = 10–1700, d_0 = 2.1–3.4 mm
Lee *et. al.* (Lee et al., 2016)	50–75% glycerol	steel, parafilm, sandpaper, glass	No heating, Re = 40–3200, We = 1–1500, d_0 = 1.8–3.3 mm
Marmanis & Thoroddsen (Marmanis and Thoroddsen, 1996)	40% ethanol, 70% isopropanol, glycerol	line paper	No heating, Re = 1100–25000, We = 100–4000, d_0 = 4.11–5.23 mm
Marston *et. al.* (Marston et al., 2010)	50% ethanol, 50% glycerol	glass powder bed	No heating, Oh = 0.0027–2.3, H = 10–1000 mm, d_0 = 1.8–2.4 mm
Qin *et. al.* (Qin et al., 2019)	10–70% glycerol	polished stainless steel	No heating, u = 0.5–4.1 m/s, Oh = 0.058–0.070, d_0 = 2.31–2.52 mm
Rioboo *et. al.* (Rioboo et al., 2002, 2001)	glycerol	glass, PVC, polymer coating, wax, alkyl ketene dimer	No heating, u = 0.78–4.1 m/s, d_0 = 0.07–4.9 mm
Scheller & Bousfield (Scheller and Bousfield, 1995)	ethanol-glycerol, glycerol	polystyrene and glass (with and without hydrophobic treatment)	No heating, Re = 19–16400, Oh = 0.002–0.58, d_0 = 0.8–4 mm
Sen *et. al.* (Sen et al., 2020)	25–75 vol% ethanol	aluminum (mirror-finish)	T_w = 150–300 °C, u = 0.7 m/s, We = 20–50
Vadillo *et. al.* (Vadillo et al., 2009)	59–85 wt% glycerol	single crystal silicon wafer withsilicon oxide layer	No heating, Re = 9–2412, We = 3.8–34.1
Vu *et. al.* (Vu et al., 2011)	50–90 wt% glycerol	plexiglass	No heating, P_a = 0.3–6 atm, u = 1–5 m/s, d_0 = 1.5–3.8 mm
Present study	2–8 vol% ethanol	polished red copper	T_w = 75–450 °C, H = 10–70 mm, We = 7–63, d_0 = 2.60–2.84 mm

lently the wettability effect, on the droplet impact without any surface modification.

1.4. Objective of the present study

Despite significant advances in single-component droplet impact, the dynamic behaviors and spreading characteristics of ethanol–water droplet impact on a heating surface receive very limited attention. Considering the complexity of the droplet impact on a heated surface as mentioned in the previous section, the impact dynamics in the literature were only applicable to some specified conditions. Furthermore, to the best of the authors' knowledge, little has been reported concerning bi-component droplets, especially ethanol–water droplets. The present experiment aims to identify the ethanol effect on droplet impact and spreading dynamics, and also to propose impact regimes maps. The study on a single droplet may serve as a first step towards the understanding of the more complicated problem of multi-droplet impact in spray cooling.

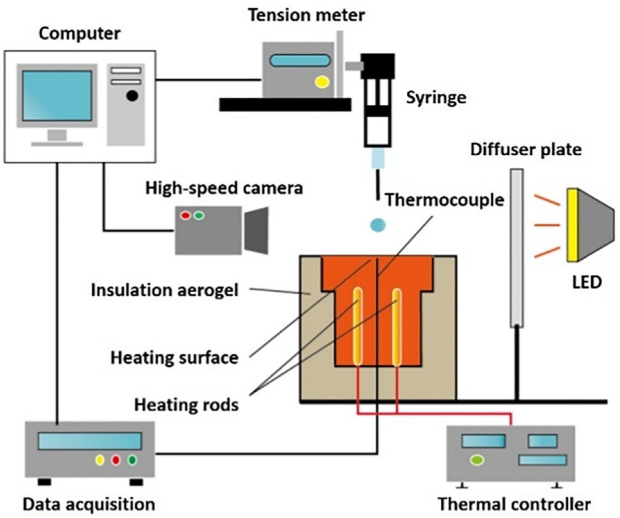


Fig. 1. Schematic of the experimental setup.

2. Experimental description

2.1. Apparatus and procedure

The droplet impact experiments are conducted by increasing the surface temperature from 75 to 450 °C under atmospheric pressure. As shown in Fig. 1, the experimental setup consists of a temperature-controlled heating system, a droplet generation system and a data acquisition system. To reduce the heat dissipation to the ambient, the heating block is enclosed peripherally with thermal insulation material (aerogel with thermal conductivity of 0.019 $W{\cdot}m^{-1}{\cdot}°C^{-1}$). Two electrical heating rods and a K-type thermocouple (Omega GG-K-30 with an accuracy of ± 0.2 °C) are vertically embedded 1 mm beneath the surface. To maintain the designed temperature, the thermocouple signal is collected and transmitted to the temperature controller to control the on/off of the heater.

A single droplet is generated at the tip of an injection syringe with an inner diameter of 0.5 mm and then falls vertically onto the horizontal heated surface. In the present experiment, the droplet is released from different heights ranging from 10 to 70 mm with an interval of 10 mm, resulting in an impact velocity range of 0.44 to 1.18 m/s. The liquid supply is computer-controlled with a predetermined volume of 8–12 μL. It needs to mention that this set volume is not exactly the actual droplet volume due to the surface tension difference caused by different ethanol concentrations. The actual volume is obtained from analyzing the high-speed images, which is shown to decrease slightly with surface tension (see Table 2).

The data acquisition instrument (Agilent 34972A) is responsible for collecting the instantaneous temperature data from the thermocouple and images from the high-speed video camera. The camera (Photron FASTCAM Mini UX100 with a spatial resolution of 1280 × 1024 pixels at a frame rate of 4000 fps) is aligned horizontally, and also normal to the generatrix of the syringe. A 120 W LED spotlight is placed opposite the camera as the illumination system.

The droplet images captured by the high-speed camera are processed by the image processing software (Image-Pro Plus 6.0 with a resolution of 0.0001 pixel). Once the pixel values of the droplet and a benchmark object are obtained, the actual droplet size can be determined by simple scale conversion. Due to several uncertain factors during the photograph, the droplet boundary captured by the software is not satisfactorily accurate. Instead, manual measurement is adopted to ensure high accuracy.

2.2. Test surface and fluids

The red copper surface used in the present experiment was initially washed with distilled water and then dried with a clean cloth. After that, it was sanded with the emery paper in a counterclockwise circumferential direction. Using Talysurf PGI 840 surface contact profilometer from Taylor Hobson, it is shown that the target surface is very smooth with arithmetic mean roughness of approximately 0.055 μm. Such a small roughness may effectively exclude the effect of surface structure on the impact dynamics and spread characteristics.

The results of spray cooling experiments by the present authors (Liu et al., 2019, 2018) showed that by adding a smaller amount of ethanol, the heat transfer coefficient increases first and then decreases after reaching a peak value, with 4 vol% ethanol aqueous solution corresponding to the best heat transfer performance. Therefore, only dilute solution with a low ethanol concentration range of 0–8 vol% is adopted as the working fluid in this paper. For the convenience of discussion, these droplets are abbreviated as E0, E2, E4, E6 and E8 corresponding to each concentration as shown in Table 2. Also tabulated in Table 2 are the thermophysical properties of water and the ethanol–water mixtures at room temperature. The surface tension and static contact angle are measured by the OSA200 Optical Surface Analyzer, while other properties for pure liquids are obtained from NIST's REFPROP 8.0 software (NIST, 2007). The properties of the binary mixures are calculated based on the recommended mixing rules (Poling et al., 2001). As can be seen in Table 2, the change in the surface tension is much more significant, *e.g.* the 8 vol% mixture shows a 30% decrease compared with pure water, and the depression is caused by the surface-active property of ethanol with respect to water. Nevertheless, the changes in all other properties by adding ethanol to water are no more than 4%.

2.3. Experimental uncertainties

Because of the high thermal conductivity of the copper surface, and also the surface is continuously heated, the uncertainty caused by the thermocouple position can be neglected. Therefore, the temperature detected by the thermocouple is regarded as the surface temperature. The error in the droplet shape analysis is estimated to be less than 2 pixels, which is equivalent to 0.02 mm with a 0.01 mm/pixel spatial resolution. As a result, the absolute uncertainty of the droplet diameter is about ± 0.04 mm. The droplet velocity is obtained by assuming free fall, *i.e.*, $u = (2gH)^{0.5}$, where the impact height H is measured by a vernier caliper with a resolution of 0.05 mm. According to the law of error propagation (Moffat, 1982), the uncertainties in the non-dimensional maximum spreading factor and droplet velocity are expressed as

$$\frac{\Delta\beta_{max}}{\beta_{max}} = \frac{\beta_{max}+1}{\beta_{max}}\frac{\Delta d_0}{d_0} \tag{1}$$

$$\frac{\Delta u}{u} = \frac{\Delta H_0}{2H_0} \tag{2}$$

which are calculated to be less than 3.02% and 0.25%.

Similarly, the uncertainty in the Weber numbers is

$$\frac{\Delta We}{We} = \frac{\Delta\sigma}{\sigma} + \frac{\Delta d_0}{d_0} + \frac{2\Delta u_0}{u_0} \tag{3}$$

Correspondingly, the maximum of the relative error is 3.12%.

3. Results and discussion

In the experiment, both pure water and ethanol aqueous solutions are adopted to study the droplet impact. One thing to note is that the Weber numbers for pure water droplets are altered by adjusting the impact velocity (or equivalently the impact height), while for the binary droplet, various Weber numbers are also achieved by changing the ethanol concentration, *i.e.*, changing the surface tension.

3.1. Phenomenological descriptions of ethanol–water droplet impact

3.1.1. Representative cases at a low surface temperature

Fig. 2 shows some typical images after droplet impact with different ethanol concentrations corresponding to the same surface temperature of 125 °C and impact height of 40 mm. The Weber number difference due to the slightly varying droplet diameter is negligibly small compared to the surface tension effect. Therefore, the former is not considered in this section.

Table 2
Physical properties of water and ethanol–water mixture at room temperature.

Concentration [vol%]		d_0 [mm]	σ [mN/m]	θ [°]	ρ_l [kg/m^3]	μ_l [mPa·s]	c_{pl} [J/kg/°C]	h_{fg} [kJ/kg]
0	E0	2.84	72.8	90.25	995.00	1.0702	4192.18	2448.92
2	E2	2.73	64.7	88.19	991.68	1.0727	4163.29	2424.72
4	E4	2.68	59.5	85.13	988.34	1.0754	4134.16	2400.32
6	E6	2.64	55.3	82.51	984.96	1.0781	4104.79	2375.72
8	E8	2.60	50.2	79.28	980.97	1.0808	4075.17	2350.92

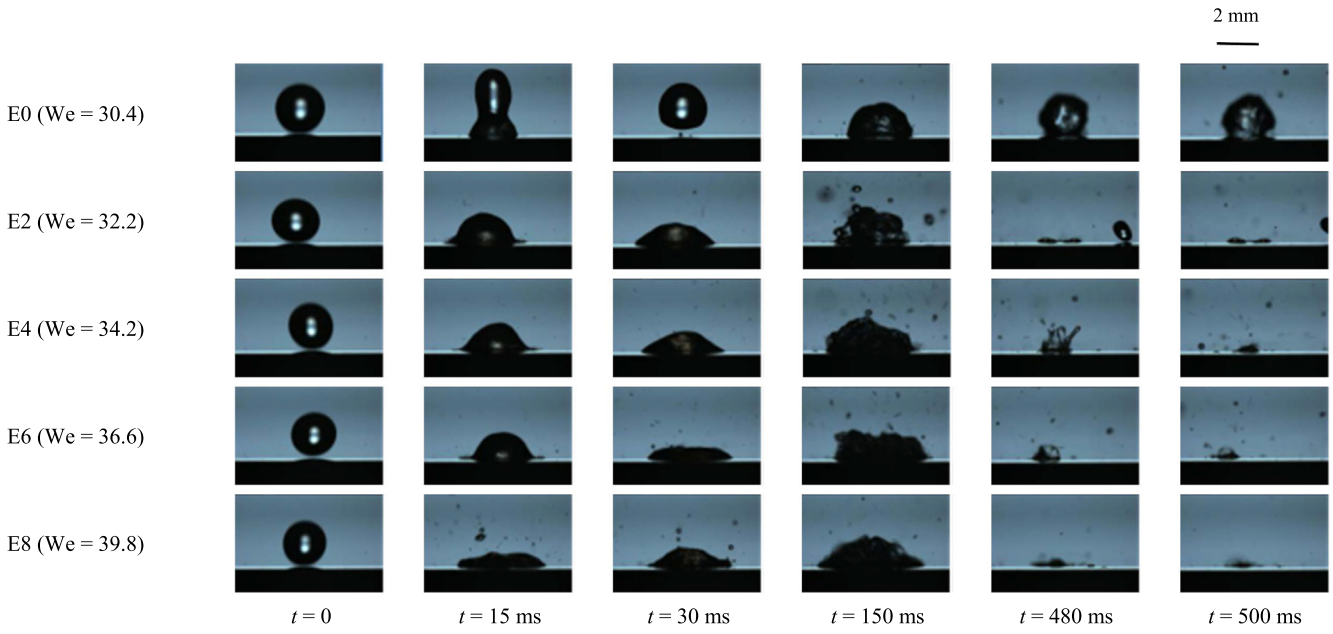


Fig. 2. Dynamic behavior evolution of different water–ethanol droplets at T_w = 125 °C and H = 40 mm (u = 0.886 m/s).

Significant differences in dynamic behaviors can be observed by adding different amounts of ethanol. It is shown that the pure water droplet undergoes impact, spread, retraction, slight rebound, second impact and boiling with minor atomization successively. By contrast, it is interesting to note that all the droplets containing ethanol deposit on the surface without rebound, and boiling occurs shortly after impact on the heated surface. It is widely acknowledged that droplets with a higher Weber number caused by higher impact velocities tend to rebound rather than deposit on the surface (Mao et al., 1997; Šikalo et al., 2002). However, the reduction in the tendency to rebound with a higher Weber number by adding ethanol is observed in the present experiment. As can be seen in Table 1, the surface tension is greatly decreased and the Weber number increases correspondingly by adding ethanol. The lower surface tension improves the spreading dynamics with the same impact kinetic energy. Because the consequent liquid film is thinner compared to that for a water droplet, viscous dissipation is also higher and inhibits droplet rebound from the heated surface. The suppression of droplet rebound by decreased surface tension is also observed in previous experiments by surfactant-added water droplets (Aytouna et al., 2010; Gatne et al., 2009). Furthermore, the binary droplets evaporate much more quickly than the water droplet, indicating an enhanced heat transfer rate. This improvement is partly related to better spreading dynamics which causes thinner liquid film. Therefore, a larger contact area is available for heat transfer. On the other hand, the ethanol additive decreases the saturated temperature of the binary droplet and increases the superheated degree, which provides more heat for bubble nucleation.

3.1.2. Representative cases at a high surface temperature

As presented in Fig. 3, the water–ethanol droplet impact behaviors are different at a higher surface temperature. For the water droplets, the Weber number is changed by varying the impact height. Only the ethanol concentration is adjusted for the binary droplets to achieve similar Weber numbers while the impact height is consistent with the corresponding water droplet. It is noted that with the increase of Weber number, atomization and breakup are more evident for pure water droplets. For example, atomization is first observed at t = 0.75 ms for the water droplet with We = 30.4, which breaks into several smaller droplets at t = 8 ms. While for water droplets with smaller Weber number of 15.2, the occurrence of atomization is delayed without droplet breakup, which is due to insufficient initial kinetic energy. It is also shown in Fig. 3 that the addition of ethanol results in violent atomization and breakup, compared with pure water droplets with similar Weber numbers. Another difference from water droplets is that no retraction occurs during the spread process of all ethanol–water droplets. This is consistent with better spreading dynamics caused by ethanol additive.

It is widely accepted that a higher Weber number is advantageous to droplet rebound and breakup, which is also confirmed by the present water results. However, some of the present results are contradictory to this general consensus. As depicted in Fig. 3, ethanol–water droplet with a lower Weber number (*e.g.*, We = 16.1 with u = 0.626 m/s) shows much stronger atomization and breakup phenomena than water droplets with higher Weber numbers of 22.8 (u = 0.767 m/s) and 30.4 (u = 0.886 m/s). This illustrates the fact that although an increased impact velocity can enhance the droplet atomization and breakup, the effect of decreased surface tension by ethanol additive is much more significant. To the best of the authors' knowledge, no similar phenomenon has been reported and no theory has been developed to explain this new phenomenon. Therefore, it might be expected that at a specified surface temperature, the Weber number is not the main factor affecting the impact outcome for the ethanol–water binary droplet. The introduction of ethanol additive can significantly improve droplet atomization and breakup, and this enhancement can even overcome the negative influence of small Weber numbers. Correspondingly, from the perspective of heat transfer, more heat can be dissipated through ethanol–water droplet impact, which is more beneficial to decrease the surface temperature during the spray cooling of electronic devices.

3.2. Effect of wall temperature on impact behavior

Since droplet impact behaviors are also highly dependent on the surface temperature, Fig. 4 is presented to illustrate the impact dynamics of pure water and E4 droplets at several typical wall temperatures. Similar to previous studies (Park and Kim, 2019;

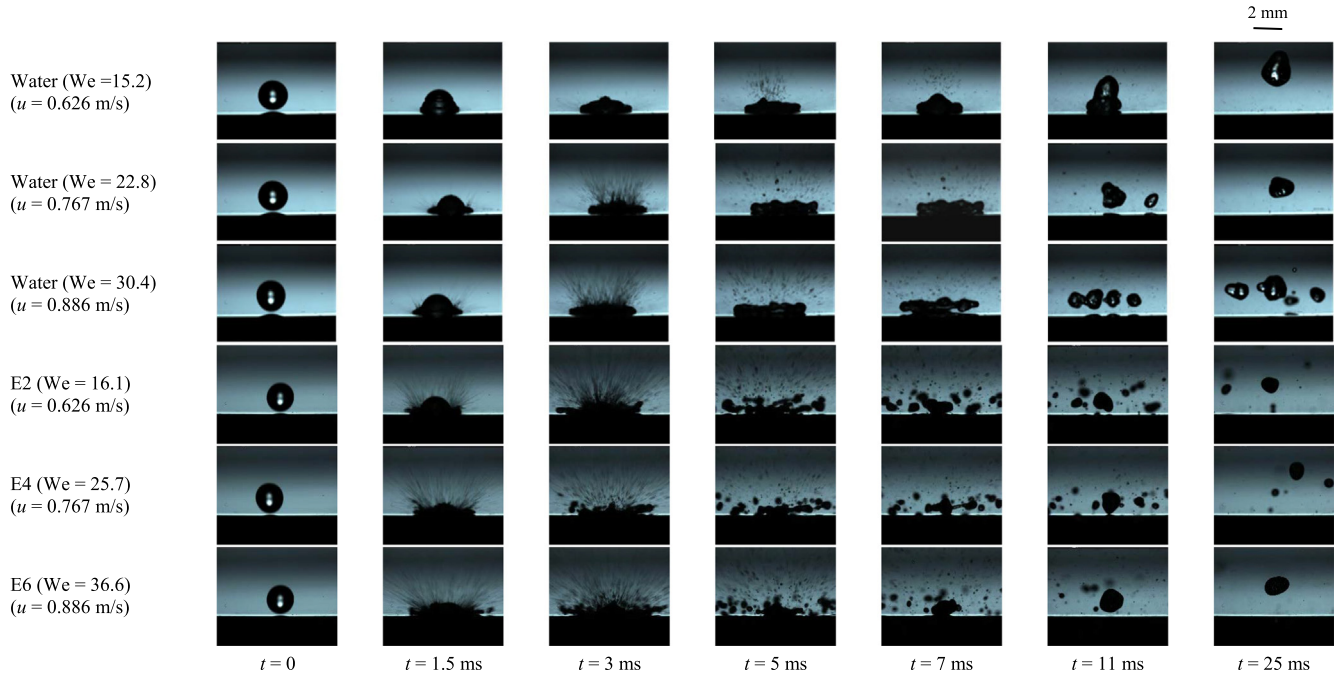


Fig. 3. Behaviors evolution of different water–ethanol droplets at T_w = 300 °C.

Wang et al., 2019), three different regions can be identified with an emphasis on the wall temperature effect based on the experimental observation, namely, film evaporation and nucleate boiling region, transition boiling region and film boiling region. Fig. 4 shows that the droplet impact behavior exhibits a complex manner in these regions.

At lower surface temperatures, droplets will deposit on the heated surface until complete evaporation without rebound or atomization. From the heat transfer perspective, this region corresponds to film evaporation and nucleate boiling. As shown in Fig. 4, both water and E4 droplets at T_w = 75 °C undergo a series of spread and retraction, reaching a steady-state at approximately t = 80 ms. At a higher surface temperature of 125 °C (see Fig. 2) which exceeds the droplet boiling point, atomization is clearly seen accompanied by violent droplet boiling for the E4 droplet. The decreased viscous dissipation losses increase the energy available for droplet spreading. Therefore, monotonically increased spreading ability is expected with the increase in surface temperature.

Droplet rebound with atomization and breakup with atomization occur during transition boiling. However, the droplet spreading property does not exhibit a monotonous variation with surface temperature. For droplet rebound with atomization at ~125–175 °C, with the increase of wall temperature, enhanced heat transfer from the surface to the droplet compensates for the energy loss caused by impact and viscous dissipation. Therefore, more kinetic energy is available for the droplet rebound accompanied by the second atomization, resulting in a weaker spread. Take the case of T_w = 150 °C as an example. E4 droplet undergoes violent atomization, and this is because it possesses a higher We (due to smaller surface tension), and correspondingly a larger contact area improves heat transfer. The decreased saturated temperature is also beneficial for nucleate boiling heat transfer. Also noted in Fig. 4 is that water droplet exhibits relatively stronger rebound ability, which restrains its spreading on the surface. Further increasing the surface temperature to ~ 175–325 °C, the absorbed energy by the droplet meets the requirement for breakup. For example, significant breakup and atomization in a radial pattern are observed at T_w = 300 °C.

For the droplet rebound during the film boiling regime, the droplet is separated from the wall by an insulating vapor film immediately after impact, which is known as the Leidenfrost phenomenon. This vapor film deteriorates heat transfer and prevents direct droplet-wall contact, making the droplets quickly rebound from the surface. Following the events depicted at T_w = 375 °C in Fig. 4, including lateral spreading, then recoil and bounce off the heated wall, little secondary droplet formation can be observed. At such high surface temperatures, the droplet spreading behavior is suppressed, and droplet impact is governed by thermal effect.

3.3. Effect of ethanol additive on spreading behavior

Fig. 5 compares the time evolution of the nondimensional wetting diameter ($\beta = d/d_0$) for droplets with different ethanol concentrations and Weber numbers at two typical surface temperatures. The maximum spreading factor $\beta_{max} = d_{max}/d_0$ is the most commonly used parameter to evaluate the droplet spread behavior, where d_{max} is the maximum diameter of the flattened area at the bottom of the droplet contacting with the heated surface. β_{max} is an indicator of the droplet spread characteristics, and a larger value means better wettability under a certain impact condition. It is shown in Fig. 5(a) that at a lower surface temperature of 125 °C, rebound only occurs for the pure water droplet with a high Weber number of 30.4, while all other droplets deposit and spread on the heated surface without any rebound even for E6 droplet with Weber number of 34.3. After reaching β_{max}, droplet oscillation is observed, but finally, the droplets tend to maintain nearly constant wetted areas indicated by the slightly fluctuant wetting parameter. Therefore, it is postulated that the liquid film thickness and contact angle decrease monotonically due to evaporation. On the other hand, Fig. 5(b) shows that a much higher surface temperature induces decreased wetting area with time until the droplets are fully evaporated in the final stage.

During droplet impact, several forces are dominant depending on the test conditions, including the capillary, inertial, and viscous forces. The relative significance of each force determines the dynamic impact behavior. Some studies proposed that when the

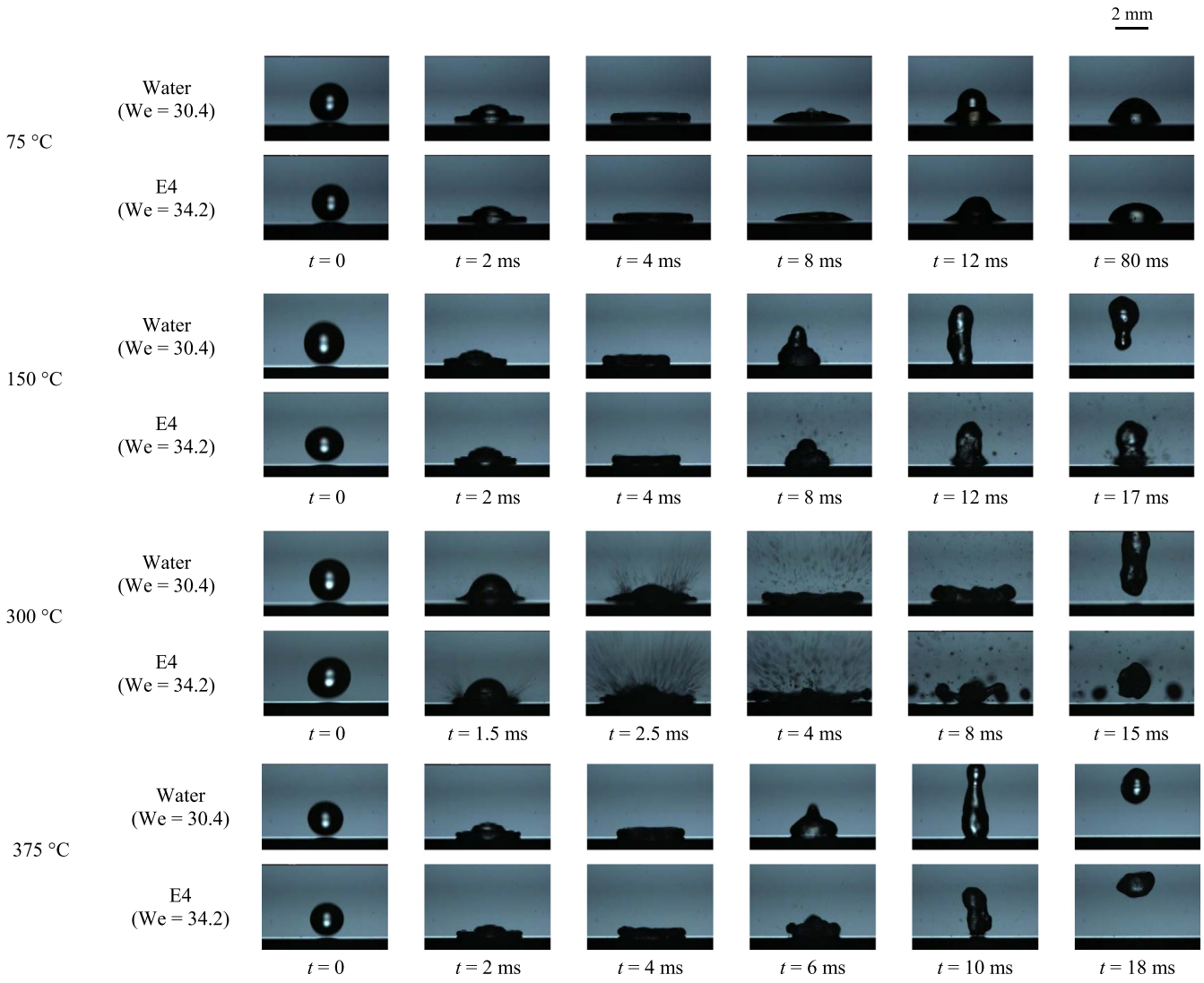


Fig. 4. Dynamic behaviors evolution of water and E4 droplets with different wall temperatures at H = 40 mm (u = 0.886 m/s).

impact factor $WeRe^{-4/5}$ is larger than unity, it can be considered as a viscous condition, and β_{max} is determined exclusively by the impact Reynolds number as $\beta_{max} \sim Re^{1/5}$. For low-viscosity liquids with $WeRe^{-4/5}$ less than 1, Re can be omitted from the correlation as $\beta_{max} \sim We^{1/4}$ (Bartolo et al., 2005; Bobinski et al., 2017; Clanet et al., 2004). In the present experiment, the impact factor in all cases is no more than 0.08, so it is hypothetically concluded from previous studies that here β_{max} is hydrodynamically determined by the Weber number, and β_{max} should increase monotonously with increased We. This is consistent with present pure water droplet results (see Table 3), which indicates that increasing the Weber number can promote the spreading and rebound possibility of water droplets. On the other hand, adding ethanol to water can decrease the droplet surface tension, inducing a higher Weber number with the same impact velocity. In this case, ethanol–water droplets also possess a higher value of β_{max}, which implies a better wettability.

However, by comparing the results of water and binary droplets, it is very interesting to note that droplet spreading dynamics doesn't seem to be solely affected by the Weber number. As can be seen from both Fig. 5(a) and Table 3, even though the Weber numbers for E2 and E4 droplets (16.1 and 25.7, respectively) are lower than that for some water droplets (We = 22.8, 30.4), the former corresponds to higher values of β_{max} and exhibit better spreading dynamics. The same phenomenon is also observed at T_w = 300 °C. For example, the E4 droplet with a smaller Weber number (We = 25.7) shows a comparable spreading dynamics with the water droplet of We = 30.4, evidenced by almost the same β_{max} values of 1.995 and 2.005.

Therefore, it can be speculated that the presence of a small amount of ethanol in water not only changes the thermodynamic and transport properties of the droplet, but more importantly, substantially changes the mechanisms during droplet impact. With the introduction of ethanol, the surface tension decreases greatly, which lowers the contact angle, and accordingly increases the wetted area and enhances the droplet spreading dynamics. In fact, since the droplet is in a non-equilibrium state during experiments, the dynamic surface tension and contact angle should be used to discuss the droplet impact. However, it is shown from the literature that generally a smaller static contact angle also corresponds to a smaller receding contact angle. Therefore, the static contact angle is used instead to qualitatively analyze the ethanol effect for simplicity. Furthermore, for an ethanol–water droplet, the preferential evaporation of the more volatile component from the droplet surface adjacent to the heated surface reduces the local ethanol concentration compared with the rest of the droplet surface. In other

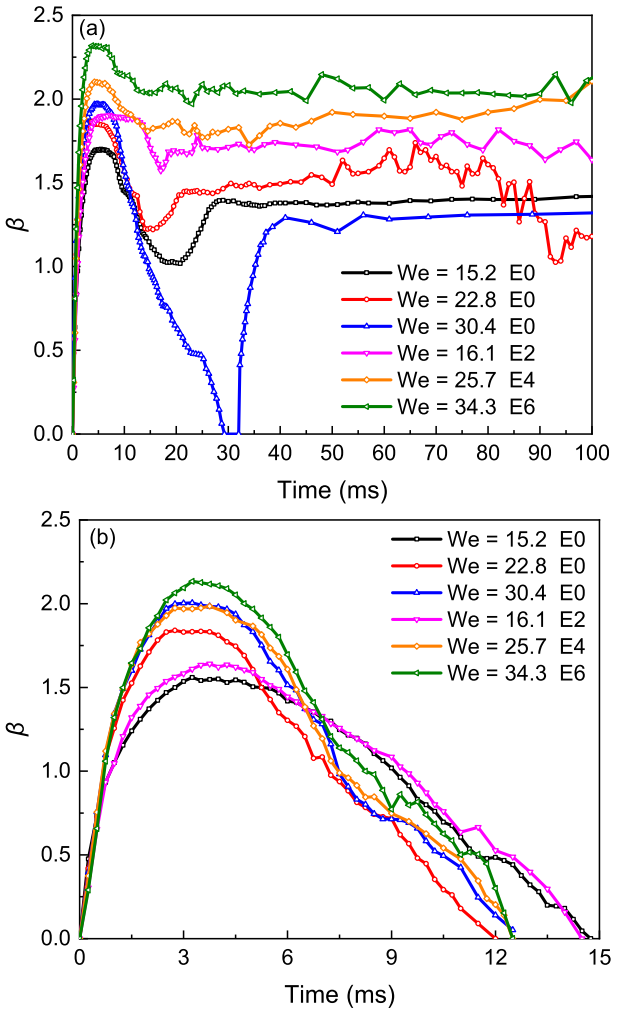


Fig. 5. Evolution of the nondimensional wetting diameter for droplets with different ethanol concentrations and Weber numbers at (a) T_w = 125 °C and (b) T_w = 300 °C.

words, the ethanol concentration near the droplet surface decreases toward the heated surface. This concentration difference is accompanied with surface tension gradient, and consequently generates a Marangoni force, pushing the droplet to flow from the low-surface-tension region to the high-surface-tension region. Therefore, the intensive Marangoni convection will improve liquid coverage of a heated surface, and this effect can be strong enough to compensate for the negative influence of small Weber numbers. Maybe this is the main reason that a binary liquid mixture with smaller Weber numbers in the present study exhibit similar or even much larger values of β_{max} compared with water droplets.

3.4. Ethanol-water droplet impact regime map

As discussed in the previous section, there are extensive studies on droplet impact regime maps with different pure liquids under various conditions. However, little attention has been drawn to the binary droplet. It is shown from the present results that the ethanol concentration significantly affects the droplet impact behavior, and consequently induces entirely different impact outcomes. Therefore, in this section, regime maps are respectively presented in Fig. 6 with a total of 339 and 411 data points for water and ethanol–water droplets impact on the heated polished red copper surface, with the wall temperature, Weber number and ethanol concentration ranging from 7 to 63, 75–450 °C and 0–8 vol%, respectively.

Fig. 6 shows the demarcation of five distinct regimes, *i.e.*, deposition, rebound, rebound with atomization, breakup with atomization and breakup. With increased surface temperature, the maps show the transition from one regime to the other at a certain critical value, which also changes with the Weber number. In addition, the maps also show a complex transition pattern with an increased Weber number at various surface temperatures.

It needs to mention that in Fig. 6b, various Weber numbers are achieved for the binary droplet by changing the ethanol concentration, *i.e.*, changing the surface tension, and also by adjusting the impact height, *i.e.*, changing the impact velocity. Series of experiments were conducted with different combinations of ethanol concentration and droplet velocity. Interestingly, the binary droplets with different ethanol concentration and droplet velocity but a nearly same Weber number show almost the same impact outcomes. Therefore, it is concluded that the respective effect of concentration and velocity can be integrated into the Weber number.

The general impact outcomes for water droplets are summarized as follows: (1) At low surface temperatures, droplets will deposit on the heated wall, regardless of the Weber number within the test range; (2) Increasing T_w designates the transition from deposition to rebound, which is also independent of Weber number; (3) At higher T_w, rebound with atomization and breakup with atomization are induced, respectively; (4) Rebound and breakup occur at low and high Weber numbers when T_w is further increased. Although the classification of these five impact regimes can also be applied to ethanol–water droplets, the ranges of surface temperature and Weber number for each regime show significant deviation. The most intuitive distinction is that the regimes of rebound with atomization, breakup with atomization and breakup are greatly extended, which shows that the ethanol additive can improve the droplet atomization and breakup. Furthermore, rebound with atomization regime shifts to the lower temperature region, and droplet rebound seems to be suppressed, both under moderate and high-temperature conditions.

Also shown in Fig. 6 are the surface temperatures at the critical heat flux and Leidenfrost point. As mentioned in Section 1.2, previous studies proposed that these two critical temperatures are considered as the limit for deposition and atomization (Bertola and Sefiane, 2005; Wang et al., 2019). It is shown that with the increase of Weber number, T_{CHF} decreases but T_{LFP} increases, indicating a broader range of transition boiling regime, in which rebound, rebound with atomization and breakup with atomization are likely to occur. The increase of T_{LFP} with Weber number has also been observed in previous experiments on various surfaces (Bertola and Sefiane, 2005; Chen et al., 2018; Misyura, 2017; Park and Kim, 2019; Wang et al., 2019; Yao and Cai, 1988). This is because

Table 3
Summary of β_{max} for various droplets in Fig. 5.

T_w = 125 °C					T_w = 300 °C				
Water		Ethanol-water		Increment	Water		Ethanol-water		Increment
We	β_{max}	We	β_{max}		We	β_{max}	We	β_{max}	
15.2	1.701	16.1 (E2)	1.899	11.64%	15.2	1.559	16.1 (E2)	1.640	5.19%
22.8	1.866	25.7 (E4)	2.326	24.65%	22.8	1.840	25.7 (E4)	1.995	8.42%
30.4	1.970	34.3 (E6)	2.425	23.10%	30.4	2.005	34.3 (E6)	2.133	6.38%

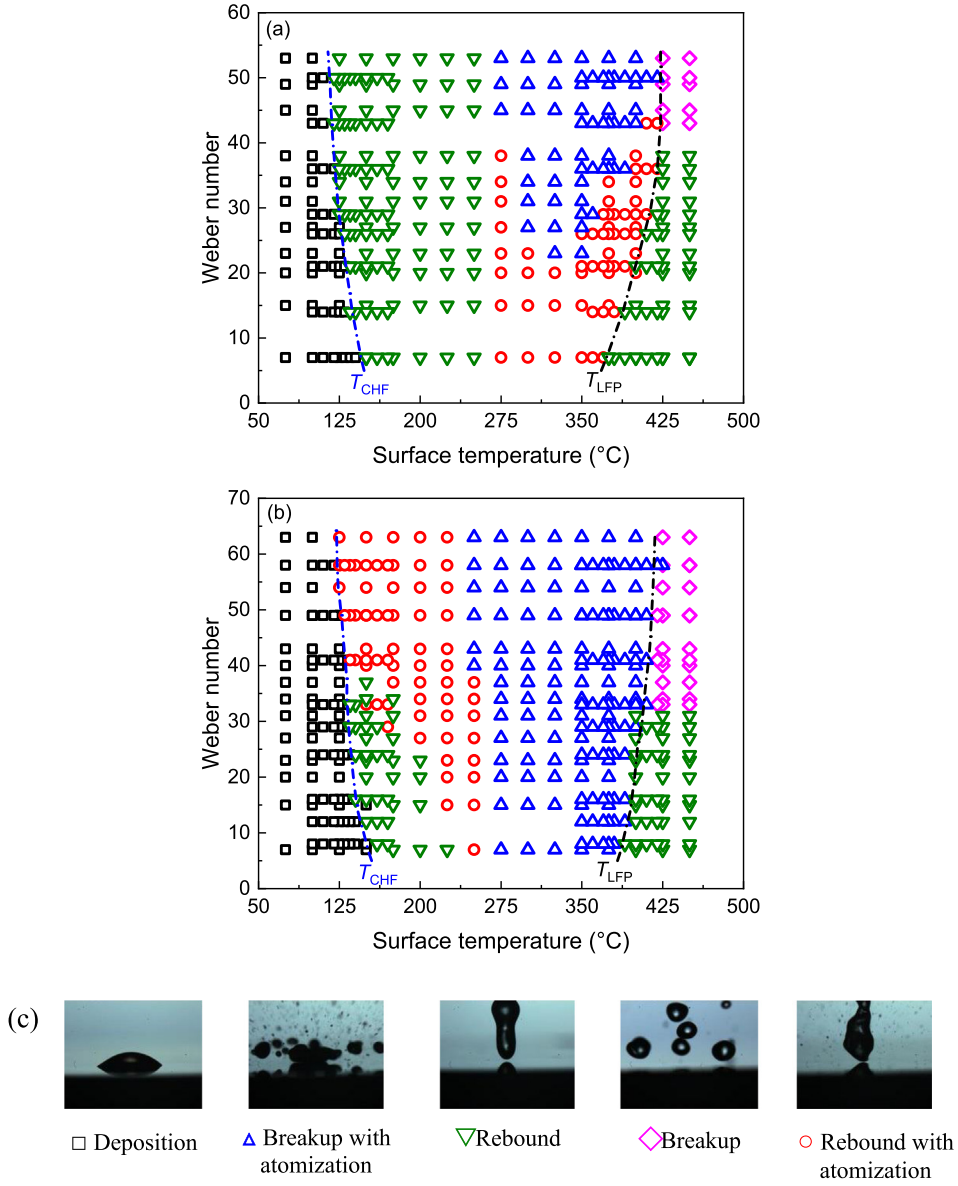


Fig. 6. Droplet impact regime map for (a) pure water droplet, (b) ethanol–water droplet, and (c) different visual features of various impact regime.

as the Weber number increases, the droplet dynamic pressure is enhanced, and consequently, the vapor film between the droplet and the heated surface can be squeezed out easily. Therefore, to maintain a continuous vapor film in the Leidenfrost region, a higher surface temperature is necessary to generate more vapor to increase the vapor pressure. On the other hand, although there is no specification of T_{CHF}, it is shown in a previous reference (Bertola, 2015) that the temperature limit for droplet deposition decreases with the Weber number, consistent with the present experimental results. T_{CHF} demarcates the nucleate boiling and transition boiling, the former of which is characterized by bubble nucleation. Note that increasing Weber number for each kind of droplet raises the droplet spread diameter. Correspondingly, a larger contact area is induced, which requires less bubble generation at a lower surface temperature to shift from droplet deposition to rebound. The determination of T_{CHF} and T_{LFP} is sort of hydrodynamic, which is different from the traditional method from the heat transfer point of view, *i.e.*, the surface temperatures at critical heat flux and Leidenfrost point correspond to the shortest and longest local droplet evaporation time.

4. Conclusions

In the present paper, experiments covering a wide range of surface temperatures and Weber numbers are conducted to investigate the dynamic behaviors and regime maps of the water and ethanol–water droplet. A high-speed camera is used to capture the instantaneous shape evolution of each droplet. The effects of ethanol additive and surface temperature on the impact and spreading dynamics are analyzed. Main conclusions can be summarized as follows:

(1) At the same impact height, the ethanol–water droplet possesses a higher Weber number due to smaller surface tension, which tends to suppress droplet rebound but to enhance droplet evaporation.

(2) A new phenomenon is observed that ethanol–water droplet with lower Weber number shows better spreading, atomization and breakup possibility compared with water droplet with higher lower Weber number.
(3) Impact regime maps are established demarcating deposition, rebound, rebound with atomization and breakup with atomization, where ethanol additive significantly changes the regime outcomes.
(4) The surface temperatures at critical heat flux and Leidenfrost point are determined from a hydrodynamic point of view. The former decreases with the Weber number while the latter increases with the Weber number.

CRediT authorship contribution statement

Hong Liu: Conceptualization, Methodology, Funding acquisition, Writing - review & editing. **Chao Si:** Investigation, Formal analysis, Writing - review & editing. **Chang Cai:** Visualization, Writing - original draft, Writing - review & editing. **Chuanqi Zhao:** Investigation, Data curation. **Hongchao Yin:** Investigation, Resources.

Declaration of Competing Interest

The authors declared that there is no conflict of interest.

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